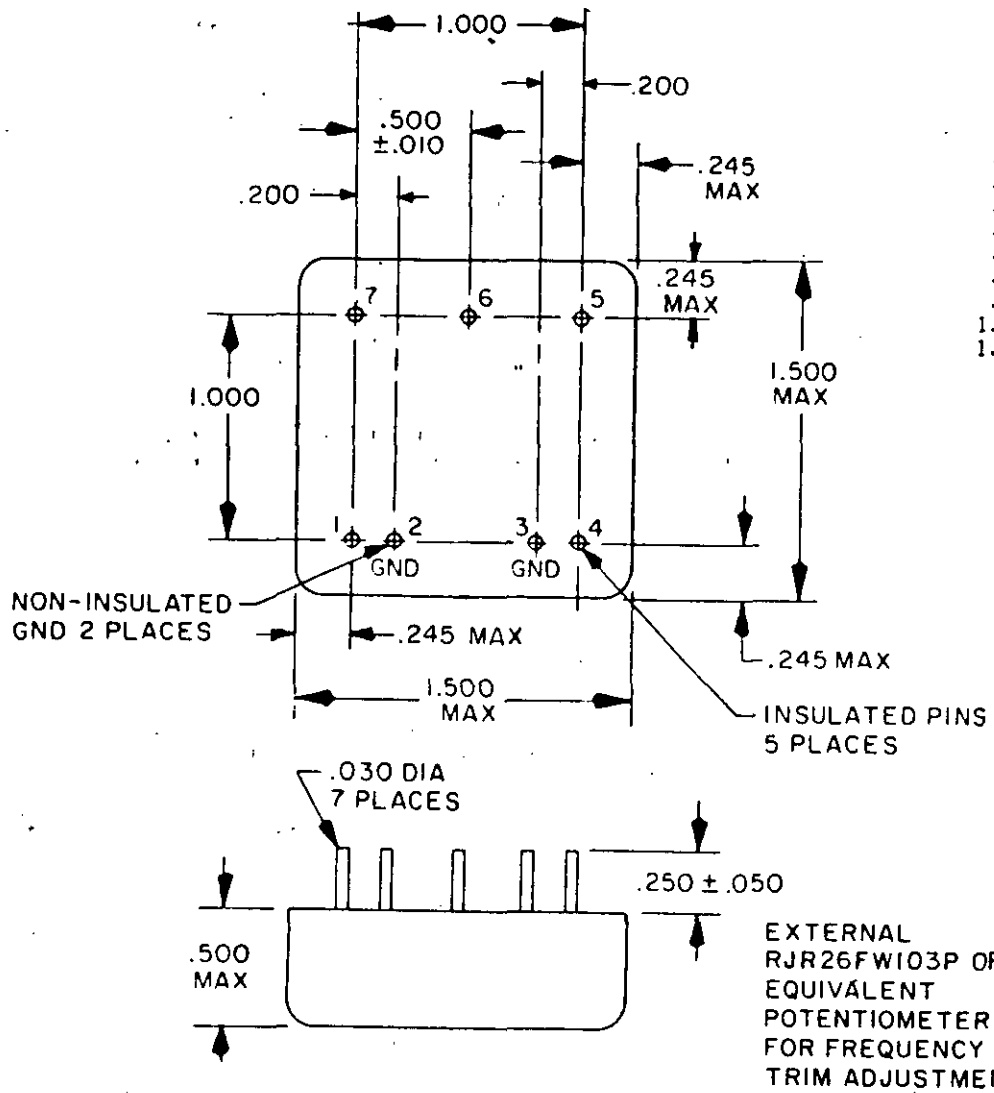
MILITARY SPECIFICATION SHEET

OSCILLATORS, CRYSTAL, CLASS 3, (TEMPERATURE COMPENSATED (TCXO)), 3.2 MHz SOLDER SEAL, SQUARE WAVE, CMOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the oscillators described herein shall consist of this specification and the latest issue of MIL-O-55310.



INCH	MM
.010	0.25
.020	0.51
.030	0.72
.050	1.27
.200	5.08
.245	6.22
.250	6.35
.500	12.70
1.000	25.40
1.500	38.10

PIN	FUNCTION
1	+10V
2	GND
3	GND
4	OUTPUT
6	TRIM
5	TRIM
7	TRIM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design and construction.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Solder seal, maximum leakage rate 1×10^{-5} atm cc/s, MIL-STD-202, method 112, test condition D.

Weight: 2.0 ounces, maximum.

Construction technology: Discrete components.

Crystal unit: Crystal unit shall be processed and tested in accordance with MIL-C-3098.

Passive discrete parts: Established reliability (ER) parts with minimum failure rate (FR) level P in accordance with MIL-STD-690. When parts of established reliability are not available QPL parts shall be used. When no QPL part is available, parts shall be tested to the applicable military specification, e.g., thermistors shall be tested to the applicable requirements of MIL-T-23648.

Semiconductor devices: JANTX in accordance with MIL-S-19500. When not available, devices will be tested and screened to equivalent JANTX requirements for similar types.

Integrated circuits: QPL parts or by DESC drawing when QPL part not available. All others shall meet vendor equivalent screening and quality conformance requirements of MIL-STD-883, method 5005, class B.

Any exception taken to selection of parts specified above shall fall into the category of non-standard parts and shall be submitted to the qualifying activity for approval.

PERFORMANCE:

Reference temperature: 21°C ±1°C, unless otherwise specified herein. Tolerance at all other temperatures, unless otherwise specified herein, shall be ±3°C.

Supply voltage: 10 V dc ±3%, peak-to-peak ripple of 50 mV at a ripple frequency of 1 Hz to 1 MHz.

Overvoltage survivability: Application of 15 V dc at the supply voltage terminals for a duration of 5 minutes shall not impair performance of the oscillator.

Input current: 5 mA maximum at 10.3 V and test circuit defined. The input current shall be determined at temperatures of -55°C, +21°C, and +85°C and at nominal load conditions following stabilization for at least 15 minutes.

Calibration tolerance: ±0.5 ppm. The output frequency of the oscillator with pins no. 5 and 6 shorted and operating at specified nominal supply voltage and load shall be measured following stabilization for at least 15 minutes at the reference temperature. The calibration error shall be determined by the difference between the measured frequency and an initial recorded frequency. Initial frequency and measurement temperature shall be recorded prior to group A and C tests and at time of shipment. The latter shall be supplied by serial number for each oscillator produced and shipped. Calibration tolerance shall be the last test in the sequence of group A tests.

Frequency adjustment: External, using a RJR26FW103P potentiometer. Minimum adjustment range shall be ±3.0 ppm of marked frequency offset at the reference temperature. Resolution shall be least ±0.1 Hz. The oscillator, operating at nominal supply voltage and load, shall be stabilized at the reference temperature for at least 15 minutes. It shall be determined by frequency measurement

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that the output signal is capable of being set to the marked frequency offset with the specified resolution and adjusted over the specified minimum trim range.

Frequency-temperature stability including hysteresis and trim effect: The frequency-temperature stability including hysteresis and trim effect for ±3.0 ppm trim shall not exceed ±2 ppm over the temperature range of -55°C to +85°C. The F-T slope at the reference temperature shall not exceed ±1 x 10^{-7}/°C. The difference between the marked frequency offset and calculated frequency offsets, including offsets determined for upper and lower trim shall not exceed 5 x 10^{-7}. Prior to start of test, the oscillator including specified frequency trim potentiometer shall be stabilized at the reference temperature and its frequency adjusted to the marked frequency offset under nominal supply voltage and load conditions. Next, the oscillator shall be stabilized at the lowest specified temperature for 30 to 120 minutes. After stabilization, the first frequency and temperature are recorded. Then the temperature is incremented in 5°C steps allowing 25 ±15 minutes stabilization at each step before data is taken until the higher specified temperature is reached. The temperature is then incrementally decreased and data acquired in a similar manner until the lowest specified temperature is reached. The temperature stability, S_T and F_{os}, shall be computed as follows:

$$S_T = \frac{\pm (F_{max} - F_{min})}{2 F_{nom}}$$

$$F_{os} = \frac{F_{RT} + F_{RT1} - F_{max} - F_{min}}{2}$$

where F_{RT} and F_{RT1} are the reference temperature frequencies for the run up and down respectively and which are determined from fitting a second order polynomial to the 15°C, 20°C, and 25°C ±2°C tolerance data points. The slope at the reference temperature shall also be computed from the fitted equation. After the initial run, the oscillator shall be stabilized at the reference temperature and the frequency trimmed a minimum ±3 ppm of the marked frequency offset. The above frequency temperature stability test procedure shall be repeated for the two trim conditions. Stability, frequency offset and slope shall be computed as above. The test may be combined into a single temperature cycle providing the frequency adjust potentiometer is adequately modeled to simulate actual conditions at the trim terminals. The potentiometer modeling method to be employed shall be approved by the qualifying activity.

Frequency-voltage stability: ±0.25 ppm for a supply voltage variation of ±3% of nominal voltage. Measurement shall be performed at temperatures of -55°C, +21°C, and +85°C and at nominal load conditions following stabilization for at least 15 minutes.

Frequency-load stability: ±0.25 ppm for a load variation of ±1.5 pF of nominal 10 pF load. Measurement shall be performed at a temperature of +21°C and at nominal supply voltage conditions following stabilization for at least 15 minutes.

Phase noise, steady state: The phase noise under quiescent conditions shall not exceed tne limits defined by the straight line segments drawn on log-log axes through the points specified in table I and shall be determined in accordance with N8S Monograph 140.

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TABLE I. Phase noise.

Phase noise measured in a 1 Hz bandwidth	
Offset from carrier	dBc
10 Hz	-80
100 Hz	-100
1 kHz	-120
10 kHz	-140

Phase noise, vibration: When subjected to vibration (operating) specified herein, the phase noise shall not be degraded more than 40 dB from the steady-state limits defined above.

Output waveform: Square wave, see figure 2.

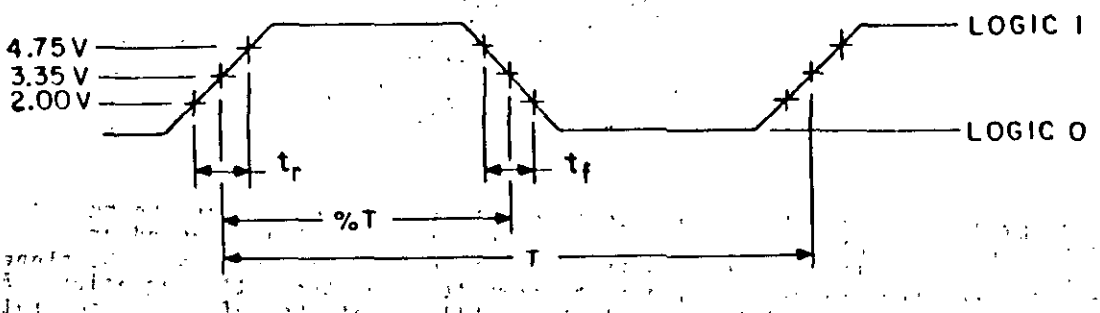


FIGURE 2. Waveform

Output voltage:

Logic 1: 6.0 V minimum, 7.0 V maximum.

Logic 0: -0.5 V minimum, 0.75 V maximum.

Measurement of output voltage shall be performed at -55°C, +21°C, and +85°C and at nominal supply voltage and load conditions.

Rise and fall times: The rise and fall times as shown on figure 2 shall not exceed 35 nanoseconds. Measurement shall be performed at -55°C, +21°C, and +85°C and at nominal supply voltage and load conditions.

Symmetry: The symmetry shall not exceed the limits of 40 to 60 percent when measured at the crossover levels specified on figure 2. Measurement shall be performed at -55°C, +21°C, and +85°C and at nominal supply voltage and load conditions.

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Spurious outputs: The nonharmonically related spurious signals shall be at least 80 dB below the main response within a 10 kHz to 100 MHz bandwidth.

Start-up time: The time for the oscillator output to achieve continuous waveform at 90% of final amplitude following application of power shall not exceed 50 milliseconds and shall be determined at temperatures of -55°C, +21°C, and +85°C and at nominal supply voltage and load conditions. An oscilloscope synchronized to the switched supply voltage or equivalent time interval method shall be employed to perform the measurements.

Frequency aging: Frequency aging at 60°C shall not exceed the following limits:

Change between any successive days	5×10^{-8}
Total change from day 0 to day 30	3×10^{-7}
Projected total change for 2.5 years	1.5×10^{-6}

The frequency aging test shall be conducted as follows: Crystal oscillators shall be maintained at 60°C ±3°C for a continuous period of 30 days. After insertion into the oven, the crystal oscillators shall be stabilized for two days at the aging temperature prior to beginning the data acquisition cycle. The time origin for data analysis shall be the beginning of the stabilization period. The frequency of each unit shall be measured immediately after the stabilization period. The frequency of each unit shall be measured a minimum of 4 times per week for 28 days at intervals of at least 20 hours. The data obtained shall be fit using the method of least squares to the function:

$$f(t) = A(\ln(Bt + 1)) + fo$$

where f(t) is the frequency of the crystal oscillator t days after the start of the aging cycle, and A, B, and fo are constants to be determined from the least square fit. The calculated frequency fo is for the beginning of the aging cycle.

If the aging trend is not monotonic, the measurement period shall be extended to 40 days after the extremum in the aging trend. For the total aging period of n days, the data for the last 28 days shall be fit to the above function. The total frequency change from day n-30 to day n and for 2.5 years (projected) shall be determined from the equation using the above determined constants. The maximum frequency shift between any two successive days shall be determined from the original data. The least square fit variance of the data from the variance shall not exceed 10% of the total aging change allowed during the test period.

Conducted interference: The conducted interference at pins no. 1 and 5 shall be a ripple not to exceed 20 mV peak-to-peak (pin 1) or 200 mv peak-to-peak (pin 5) in a 10 Hz to 100 MHz bandwidth. Using the load test circuit of figure 3, the conducted interference shall be measured using an oscilloscope having a minimum bandwidth of 10 Hz to 100 MHz. A bypass capacitor having a minimum value of 1,000 pF shall be used on the dc input terminal to ground. Length of leads between potentiometer terminals and pins no. 5, 6, and 7 shall not exceed 1.5 inches. Conducted interference at pin no. 5 shall be measured with the potentiometer set approximately in the center and at the two extremes of its adjustment range.

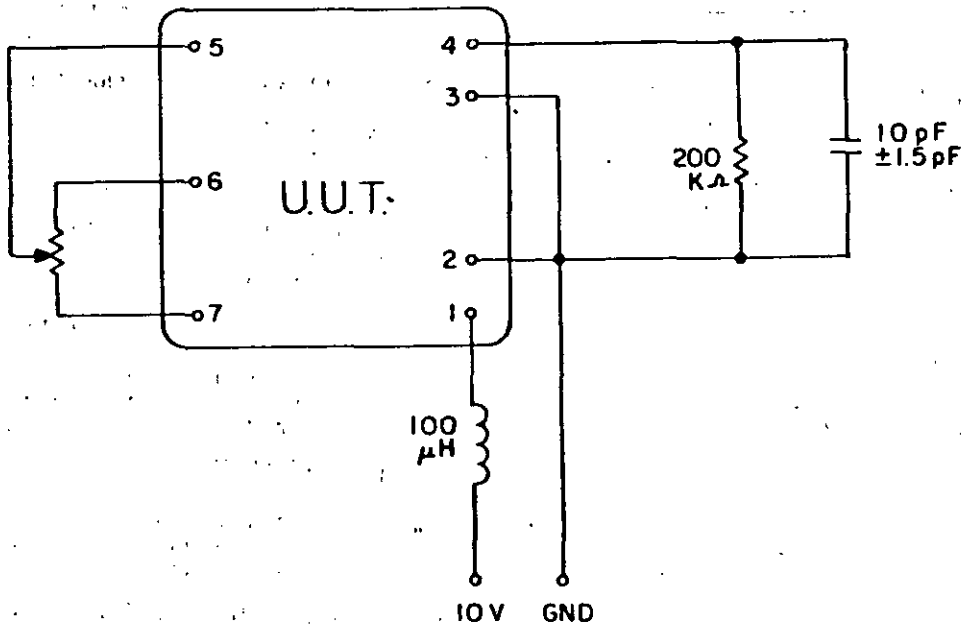
Terminal strength: MIL-STD-202, method 211, test condition A, applied force - 2 pounds.

Vibration (nonoperating): MIL-STD-202, method 204, test condition A, except upper frequency shall be 2,000 Hz, sweep time 30 minutes, and test duration 18 hours.

Vibration (operating): The specimen, mounted as specified in MIL-O-55310 and operating under nominal supply voltage and load conditions, shall be subjected to a sinusoidal vibration amplitude as follows:

Frequency	Amplitude
10 Hz - 40 Hz	.06 in. DA ±10%
40 Hz to 2000 Hz	5g (peak ±10%

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NOTES:
1. Component parts or equivalent.
2. When pins 5 and 6 are shorted, frequency shall increase.

FIGURE 3. Load test circuit.

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The vibration frequency shall be advanced from 10 Hz to 2,000 Hz in a logarithmic progression allowing no fewer than 10 phase noise measurements per decade (i.e., $10^{1.0}$, $10^{1.1}$, $10^{1.2}$, ... $10^{3.3}$) and shall be performed once in each of three mutually perpendicular directions. Total vibration duration shall not exceed 120 minutes (40 minutes each direction). The phase noise under vibration requirement shall be met.

Thermal shock: MIL-STD-202, method 107, test condition B-1.

Altitude (operating): MIL-STD-202, method 105, test condition B. The calibration tolerance requirement shall be met.

Resistance to soldering heat: MIL-STD-202, method 210, test condition A except, temperature 315°C ±7°C, immersion duration 30 seconds, immersion depth within ±0.05 ±0.005 inches of case, cooling time prior to measurement 30 minutes.

Storage temperature: -62°C to +125°C.

Resistance to solvents: MIL-STD-202, method 215.

Marking: In accordance with method I of MIL-STD-1285 and shall include the military type number, source code, serial number, date code, frequency offset at reference temperature (rounded to the nearest tenth of a hertz), nominal value of the output frequency (rounded to the nearest hertz), and terminal (pin) numbers. The frequency offset marking shall be as follows: ±X.X Hz at 21°C. The nominal frequency marking shall occupy a maximum of 11 spaces, as follows: A maximum of 7 digits; a space for decimal point and 3 spaces for MHz; i.e., 3.2 MHz.

QUALITY ASSURANCE PROVISIONS:

Quality assurance provision of MIL-O-55310 shall be applicable with the following exceptions.

Inspection of product for delivery: Lots shall not be delivered until sample units selected from these lots have successfully completed Group A and Group B inspection.

Group A inspection: Group A inspection shall consist of the inspections in table II.

TABLE II. Group A inspection.

Inspections
Visual and mechanical
Seal
Supply voltage
Overvoltage survivability
Input current
Frequency adjustment
Frequency-temperature stability including hysteresis and trim effect
Frequency-voltage stability
Frequency-load stability
Phase noise, steady state
Output waveform
Output voltage
Rise and fall time
Symmetry
Spurious outputs
Start-up time
Calibration tolerance

Group B inspection: Group B inspection shall consist of the frequency aging test as specified herein.

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Group C inspection: Group C inspection shall consist of the inspections in table III, in the order shown. Measurements as specified below shall be made before and after group I testing in lieu of measurements following each test. Final measurements shall then be similarly made at the end of group II and group III testing as specified below. The following requirements shall be met:

Seal
Calibration tolerance
Frequency-temperature stability including hysteresis and trim effect
Start-up time
Output voltage
Input current

Screening: All oscillators to be delivered shall be subjected to the screening tests specified in table IV in the order shown prior to any conformance inspection.

Failures: Oscillators which fail in any test criteria in the screening sequence shall be removed from the lot at the time of observation or immediately at the conclusion of the test in which the failure was observed. Rejected units may be reworked and subjected to 100% screening.

Lots resubmitted for screening: Lots may be resubmitted for screening one time only and may be resubmitted only when the observed percentage of defectives does not exceed 20%.

Qualification inspection: Test results shall demonstrate that at least 8 oscillators of the type for which qualification is requested have met the requirements for group A, B, and C inspections and a destructive physical analysis performed. The destructive physical analysis shall be performed on group II units following group C inspection acceptance. The analysis shall consist of an examination of disassembled oscillators to verify that material, design and construction, and workmanship are in accordance with the applicable requirements.

TABLE III. Group C inspection.

Inspections
Group I - all sample units
Conducted interference
Terminal strength
Vibration (nonoperating)
Vibration (operating)
Shock
Solderability
Group II - 1/2 of sample units
Thermal shock
Altitude (operating)
Moisture resistance
Fungus
Group III - 1/2 of sample units
Storage temperature
Salt spray
Resistance to soldering heat
Resistance to solvents

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TABLE IV. Screening test (100%).

Screening tests	Method/condition
	Class B
Internal visual	Soldering: MIL-S-45743, Quality conformance
	Workmanship: Per MIL-O-55310
Thermal shock	MIL-STD-202, method 107, condition B
Seal (gross)	MIL-STD-202, method 112, condition D
Burn-in 1/	192 hours at 85°C minimum
Serialization	Marking: as specified herein
Electrical tests	Frequency temperature stability including hysteresis and trim effect
	Output voltage 2/
	Input current 2/
	Start-up time 2/

1/ Under load, connected as shown on figure 3.
2/ Measurement at reference temperature ±3°C.

DEFINITIONS:

Nominal frequency: The frequency assigned to the oscillator when operated under specified conditions.

Frequency offset: The frequency difference, positive or negative, which should be added to the specified nominal frequency of the oscillator, when adjusting the oscillator frequency at specified reference temperature and operating conditions in order to minimize its deviation from nominal frequency over the operating temperature range.

Frequency temperature (F-T) stability: The total frequency change which occurs over the entire operating temperature range and is expressed as ± one-half of the total, with no reference implied (i.e., 4 ppm peak-to-peak over the entire operating temperature range is expressed as ±2 ppm).

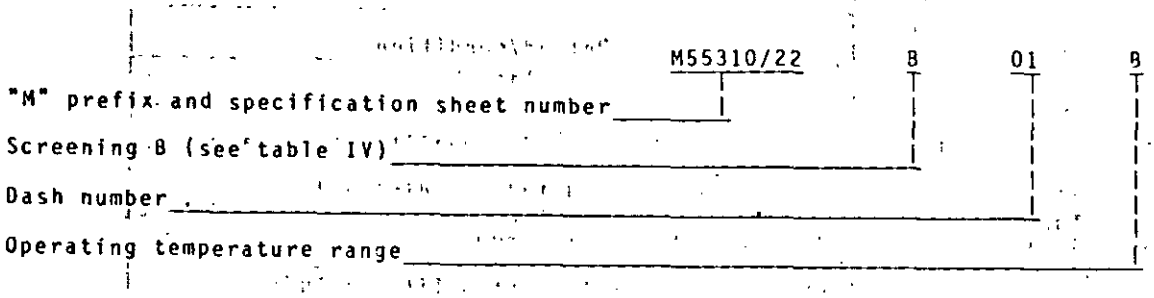
Hysteresis: A thermally induced effect in crystal resonators and crystal oscillators which manifests itself as nonrepeatability of the F-T characteristics, observed over a complete quasi-static temperature cycle.

Trim effect: The degradation of crystal oscillator F-T stability and marked frequency offset as a result of frequency trim which produces a rotation and/or distortion of the initial compensated F-T characteristic.

Calibration tolerance: The maximum permissible deviation of the oscillator frequency from an initial calibration setpoint, at specified reference temperature and nominal supply voltage and load conditions, due to all other operating parameters exclusive of time. For oscillators requiring an external frequency adjust potentiometer which is not to be supplied by the manufacturer as an integral part of the oscillator, the calibration tolerance is referenced to an initial recorded frequency obtained under specified conditions such as shorted pins at the frequency adjust terminals.

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Military part number: Consists of "M" prefix, specification sheet number, dash and coded alphas, and numeric number as shown in the following sample:



NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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